Null boxes for the Fan–Sun amended conjecture on near-tight speed tuples: a partial recomputation with an independent complete-witness screen

(no counterexample; a null box proves nothing outside the box)

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AI-assistance disclosure and audit status. This preprint, its code, its claim ledger and the audit were produced by AI agents (Claude, Anthropic) under the author's direction; the author is responsible for the content. **This version was audited by a separate AI agent instance (`AUDIT.md`, ACCEPT_WITH_CORRECTIONS), not by a human.** The audit's eight corrections are applied (Section 5). No human audit exists. There is no Lean in this paper.

Abstract

For positive integer speeds $v_1, \dots, v_n$ let ML be the lonely-runner loneliness, $\mathrm{ML} = \max_t \min_i \|tv_i\|$. A tuple is *near-tight* if $\mathrm{ML} < 1/n$. Fan and Sun's amended spectrum conjecture (Conjecture 1.3) says that $\mathrm{ML} = s/(ns+k)$ with $1 \le k \le n$, or $\mathrm{ML} \ge 1/n$; for $\mathrm{ML} = p/q$ in lowest terms and $k' = q - np$ this is exactly $1 \le k' \le n$. We enumerated all primitive tuples of distinct speeds in boxes with exact integer witnesses and recomputed ML of every near-tight tuple with Python fractions. **No counterexample was found**: no tuple has $k > n$, and none has $k > n/2$ (Fan–Sun Conjecture 6.1). The boxes 7 speeds ≤ 70 and 8 speeds ≤ 60 were already covered by Fan and Sun's own Table 1, so they are recomputations. Relative to the literature only 7 speeds in $(70, 200]$ and 8 speeds in $(60, 130]$ are new, and they are **null boxes**: 55 primitive near-tight septuples ≤ 200 ($k = 1$: 51, $k = 2$: 4), 65 octuples ≤ 130 (all $k = 1$). An independent audit (Rust, a complete witness set containing sums, differences and peaks) and a run of Cordella's own `survey.c` reproduced every tuple set exactly. **Not a proof, nothing is solved, Tier [C] throughout.**

What this is NOT. Not a proof or refutation of Conjecture 1.3; not a statement about speeds beyond the boxes; not a new finding about the septuple $(2, 7, 9, 11, 13, 20, 54)$ (Section 3); the box $8 \le 150$ preregistered in the addendum was *not run*; not formalised in Lean; not human-audited.

Markers. [L]: a cited source, taken as stated. [C]: executed exact integer/fraction code with the argument written down, not machine-checked. [X]: floating-point or recorded-absence evidence. Identifiers `LRS-·-nnnn` refer to `ledger/ledger.json`.

1 Statement, definitions, scope

$\mathrm{ML}(v) = \max_t \min_i \|tv_i\|$, $\|\cdot\|$ the distance to the nearest integer. Near-tight: n distinct positive integers with $\mathrm{ML} < 1/n$ (Cordella [2], §2.1). Write $\mathrm{ML} = p/q$ in lowest terms, $k' = q - np \ge 1$ for near-tight tuples.

Fan–Sun Conjecture 1.3 ([L], [1]). For positive integers $v_1, \dots, v_n$, either $\mathrm{ML} = s/(ns+k)$ for some $s, k \in \mathbb{N}$ with $k \le n$, or $\mathrm{ML} \ge 1/n$. With $s = gp$, $ns + k = gq$ one gets $k = gk' \ge k'$, so a representation with $k \le n$ exists iff $1 \le k' \le n$ (the audit checked this equivalence). **Conjecture 6.1** is the same with $k \le n/2$.

Fan–Sun's Table 1 (their computer search, speeds up to 70 for $n = 7$ with $k \in \{1, 2\}$, up to 60 for $n = 8$ with $k = 1$) calls $n = 7, 8$ "unconvincing due to the low search bounds"; their code is unpublished and their counts are not published.

Scope. *Distinct speeds suffice:* a tuple with a repeated speed has at most $n - 1$ distinct speeds, and then ML $\geq 1/n$ by the proved lonely runner conjecture for at most 8 runners (classical for ≤ 7 runners; Rosenfeld [4] for 8), since here $n - 1 \leq 7$ speeds. *Primitive tuples suffice:* ML is invariant under dividing out a common factor.

2 Method

Completeness of the sum-time witness set ([C]). Let $f(t) = \min_i \|tv_i\|$, $D = v_i + v_j$. Claim: if ML $= f(t_0) > 0$, then t_0 is of the form $m/(v_i + v_j)$ with $1 \leq m \leq D/2$ after the symmetry $t \mapsto 1 - t$. At a maximiser the right derivative of f is ≤ 0, so some active runner r falls to the right (value $b - t_0 v_r$, $b \in \mathbb{Z}$); the left derivative is ≥ 0, so some active runner s rises on the left (value $t_0 v_s - a$). If $r \neq s$ then $t_0 = (a+b)/(v_r + v_s)$. If $r = s$ then $t_0 v_r \equiv 1/2$, $f(t_0) = 1/2$ and every active runner sits at $1/2$, so $t_0(v_i + v_j) \in \mathbb{Z}$ for two active runners: t_0 is a sum time with $m = D/2$. Cordella's Lemma 2.2 uses sums *and* differences $m/(v_i - ev_j)$, $e = \pm 1$; for positive speeds the sums suffice, as just shown. Hence "tuple is near-tight iff no sum time $m/(v_i + v_j)$, $m \leq D/2$, has all runners at distance $\geq 1/n$" is exact, and each rejection is an exact integer witness.

Scan. `data/code/lrspec_scan.c.txt` (C, bitmask per speed over the witness list, two witness lists W4/W9 as a consistency device, exact second stage scanning $m = 1..D/2$), driven by `run_box.py`; every near-tight tuple is recomputed with Python `Fraction` (`verify.py`). Every box enumerates exactly $\binom{N}{n}$ tuples (checked). Cores 6, 7 only. *Correlated check:* `verify.py` shares the sums-only reduction with the C exact stage, so its "0 mismatches" cannot detect an error in that reduction; the auditor's screen below does not share it.

Controls (run before the headline boxes). Positive: $n = 4 \leq 200$: 4125 primitive near-tight tuples ($k = 1$: 3836, $k = 2$: 289, containing $(1, 3, 4, 14)$ with ML $= 4/17$); $6 \leq 110$: 348 (315, $k = 3$: 33); $7 \leq 50$: 29 (26; $k = 2$: three tuples); $8 \leq 40$: 29 (all $k = 1$); Fan–Sun examples 7/30, 8/51, 3/23, 5/37 and the family $(2s + 7)/(8s + 30)$, $s = 0..20$. Negative: strictness corrupted ($nd > b$) changes the counts (4125 $\to$ 14197, $n = 6 \leq 60$: 118 $\to$ 908, $7 \leq 50$: 29 $\to$ 382, $8 \leq 40$: 29 $\to$ 394), truncated witnesses ($m \leq D/4$) change them $(4143, 119, 39, 31)$, a tampered p/q is flagged. The C scan agrees with a pure-Python brute force on $4 \leq 30$, $5 \leq 24$, $6 \leq 20$, $7 \leq 18$, $8 \leq 16$, and the two witness lists give byte-identical output on $4 \leq 200$, $5 \leq 100$, $6 \leq 110$, $7 \leq 60$, $8 \leq 45$ and on $7 \leq 80$, $8 \leq 60$, $7 \leq 120$.

3 Results

Novelty, exactly. The boxes $7 \leq 70$ and $8 \leq 60$ are covered by Fan–Sun's Table 1, so they are recomputations; the 7-speed boxes ≤ 50 and 8-speed ≤ 40 are Cordella's. New relative to the literature are only 7 speeds in $(70, 200]$ and 8 speeds in $(60, 130]$, and they are null boxes. Fan–Sun's counts are not published, so ours are the first exact published counts at those bounds, and that is all they are. A same-day search (alphaXiv, 2026-11-01) found no later work extending the 7- or 8-speed spectrum searches and no refutation or proof for $n \geq 5$; Hu–Jing [3] (uniform ML gap) is not about the spectrum.

The four $k = 2$ septuples. The same four appear in every box from ≤ 54 up to 200: $(1, 2, 3, 4, 5, 7, 18)$ and $(1, 3, 4, 5, 7, 13, 18)$ with ML $= 3/23$, $(1, 3, 4, 5, 7, 11, 30)$ with 5/37, $(2, 7, 9, 11, 13, 20, 54)$ with 9/65. All were re-verified over the complete Lemma 2.2 set (optimal $t = 4/23$, $4/23$, $6/37$, $11/65 = 11/(11 + 54)$). **The last is not a new finding.** It is printed in neither paper. It lies outside Cordella's 7-speed box (maximum speed $54 > 50$), and Cordella's own `survey.c` finds it (9/65, $k = 2$) once the box reaches 54. It lies inside Fan–Sun's searched range (≤ 70), where they report $k \in \{1, 2\}$ without listing tuples. That they saw it is an inference: their search code is unpublished.

n	speeds $\leq$	near-tight	k counts	$k > n$	core-s
4	200	4125	1:3836, 2:289	0	(control)
6	110	348	1:315, 3:33	0	(control)
7	50	29	1:26, 2:3	0	(control)
8	40	29	1:29	0	(control)
7	80 (primary)	37	1:33, 2:4	0	14
8	60 (primary)	36	1:36	0	13
7	100	40	1:36, 2:4	0	66
8	80	47	1:47	0	128
7	120	44	1:40, 2:4	0	243
8	100	53	1:53	0	831
7	150	48	1:44, 2:4	0	1225
7	180	52	1:48, 2:4	0	4054
8	130	65	1:65	0	8136
7	200	55	1:51, 2:4	0	8409
6	150	627	1:562, 3:65	0	53
6	200	1085	1:971, 3:114	0	301

Table 1: Primitive near-tight tuples of distinct positive speeds ([C]). No tuple has $k > n$, and none has $k > n/2$, in any box. Core-s is 2× wall, an upper bound; the boxes (below the controls) sum to 23,473 core-s = 6.5 core-h, excluding controls and verification.

Secondary (descriptive). All 1085 sextuples ≤ 200 have ML $= (P-1)/(6P)$ with odd q ($P \equiv \pm 1 \bmod 6$), consistent with Cordella's odd-denominator theme (his Conjecture 6.2 is the odd-denominator statement, $k \in \{1, 3\}$). Maximum $v_{\max}/q$: $195/73 < 3$ for six speeds, attained at $(6, 61, 67, 128, 189, 195)$, a member of the family $(6, 31, 37, 68, 99, 105)$, $(6, 43, 49, 92, 135, 141)$, $(6, 61, 67, 128, 189, 195)$ that looks like the extremal family for which Cordella's Theorem 5.3 already proves $v_{\max} < 3q$; this is consistent with a theorem and is not new evidence about sporadic tuples. For 7 and 8 speeds the maxima are $13/5$ at $(1, 4, 5, 6, 7, 11, 39)$ and $21/17$ at $(1, 5, 6, 7, 8, 11, 13, 21)$.

4 The independent audit

A separate AI agent instance (not a person; `AUDIT.md`, ACCEPT_WITH_CORRECTIONS) wrote a screen in Rust not derived from the C code. Its witness set is *every* reduced fraction $a/b \leq 1/2$ with $b \leq 2N$ (3903 fractions for $N = 80$, 24339 for $N = 200$), which contains all sums, differences and peaks, so near-tight $\iff$ the AND of per-speed bitsets is 0, with no second stage. A faster prefilter mode (first 6 words) matched the full mode on $4 \leq 200$, $6 \leq 110$, $7 \leq 80$, $8 \leq 60$. Tuple sets were compared as lines "$v_1..v_n$ p q k": identical to the producer on every box of Table 1, including $8 \leq 130$ (1,624,177,854,000 tuples, 4133 s) and $7 \leq 200$ (2,283,896,214,600 tuples, 5339 s), and on the sub-boxes obtained by restricting. The auditor rebuilt `lrspec_scan.c` from source and reproduced $7 \leq 80$, $8 \leq 60$ byte for byte. Exact fraction checks over the complete set: all rows of $7 \leq 80$ and $8 \leq 60$ and 40 random rows of each of $7 \leq 200$, $8 \leq 130$, $6 \leq 200$, 0 mismatches. The auditor's own strictness-corruption control gives the same counts $(14197, 908, 382, 394)$, every added row having ML $= 1/n$ exactly.

Cordella's code. The auditor reviewed `survey.c` (v1.0.2, MIT; printf/OpenMP only, no file, network or exec calls) before building and running it: its tuple sets for $4 \leq 200$, $6 \leq 110$, $7 \leq 50$, $8 \leq 40$ are identical to the producer's controls, including p, q, k; `survey 56 7` gives 32 tuples ($k = 1$: 28, $k = 2$: 4), identical to the auditor's Rust $7 \leq 56$. The archive is not shipped.

5 Audit corrections applied

The producer's `data/RESULT.md`, `data/PREREG_*` and `data/result.json` are unmodified copies (pre-audit, hashed); this paper supersedes them. (1) The $(2, 7, 9, 11, 13, 20, 54)$ sentence is replaced as in Section 3. (2)

The `survey.c` run is reported (Section 4). (3) The producer's two-witness-list check on $8 \le 130$ and $7 \le 200$ was *not done*; it is superseded by the auditor's complete screen. (4) The scope sentences are added (Section 1). (5) The cost is stated as 23,473 core-s = 6.5 core-h; the unrun box $8 \le 150$ ($\binom{150}{8} = 5.26 \cdot 10^{12}$ tuples) would have added about 26,000 core-s, far below the 60 core-h cap, so skipping it was a choice and is disclosed. (6) The docstring of `data/code/ladder2.py` now reads "8<=150 not run; 7<=200 was run separately" (this is the only edited file; the lane copy is unchanged). (7) The correlated-check note is added (Section 2). (8) The `PREREG_ITEM.md` header says "written ~23:20 UTC" but its mtime is 23:16:30, before the scan binaries (23:17:09); the preregistration timing is self-attested, since the whole directory arrived in one commit, so the hashes prove content, not time.

6 Limitations and what is not done

- Null boxes only. Conjecture 1.3 is neither proved nor refuted. Both screens enumerate only the stated boxes.
- $8 \le 150$ was never run, by anyone. The producer's W9 run on $8 \le 130$ and $7 \le 200$ was not done (superseded).
- The producer's N2 (truncated candidates) and N3 (sampling) controls and the float-grid check were not re-implemented by the auditor.
- The claim that Fan–Sun likely saw the septuple is an inference. AI audit only; no human audit. Tier [C]; no Lean.
- Preregistration deviations: the optional run of Cordella's code was refused by the producer's sandbox (done by the auditor); control N3 wording corrected in addendum 1; the second ladder (addendum 1, sha256 `b6d9ebf0...7669`) was partly run; `PREREG_ITEM.md` sha256 `7e62a74e...7ac7a6`.

7 Claim ledger and reproducibility

`ledger/build_ledger.py` files the claims with hashed evidence; the Elenchus gate (`ledger/run_gate.py`, verbatim output in `ledger/gate_output.txt`) checks bookkeeping only and licenses nothing. Reproduce: `run_box.py n N label 4` (compile `lrspec_scan.c.txt` as C), `verify.py`; auditor: Rust sources in `data/audit/rs`, `fracverify.py`, `cmp.py`. Shell scripts are stored as `.sh.txt`, logs as `.log.txt`, binaries and third-party PDFs and code are not shipped. Inputs are copies of `results/proposal5/2026-10-27/LR-SPEC/` and of the auditor's scratch directory; nothing was committed, pushed or uploaded.

References

[1] H. T. Fan, A. Sun, *Amending the Lonely Runner Spectrum Conjecture*, arXiv:2306.10417 (v2, 30 Jan 2026); Electron. J. Combin. 33(1) (2026) P1.38. ([L])

[2] F. Cordella, *Odd denominators in the Lonely Runner spectrum for six speeds*, arXiv:2609.03444 (v2, Sept 2026); code `survey.c` v1.0.2, doi:10.5281/zenodo.22345079 (MIT). ([L])

[3] X. Hu, Y. Jing, *Near optimal gap in the Lonely Runner Conjecture*, arXiv:2610.23790 (16 Oct 2026). ([L])

[4] M. Rosenfeld, *The lonely runner conjecture holds for eight runners*, arXiv:2509.14111 (v2, Oct 2025). ([L])